1 Geometry of $\mathbb{C}^n$

Definition 1.1. : The set $\mathbb{C}^n := \mathbb{R}^n + i\mathbb{R}^n$ is the n-dimensional complex vector space consisting of all vectors $z = x + iy$, where $x, y \in \mathbb{R}^n$ and i is the imaginary unit satisfying $i^2 = -1$.

$\mathbb{C}^n$ is endowed with the Euclidian inner product

$$\langle z, w\rangle := \sum_{i=1}^{n} z_i \overline{w}_i \quad (1)$$

where $z = (z_1, z_2, \dots, z_n) \in \mathbb{C}^n$ and $w = (w_1, w_2, \dots, w_n) \in \mathbb{C}^n$ and the notation $\overline{z}_i$ is the complex conjugate of $z_i \in \mathbb{C}$.

$\mathbb{C}^n$ is endowed with the Euclidian norm

$$\|z\|_2 := \sqrt{\langle z, z\rangle} \quad (2)$$

$\mathbb{C}^n$ endowed with the inner product (1) is a complex Hilbert space and the mapping

$$\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}^n$$

$$(x, y) \mapsto x + iy$$

is an siometry. Hence all metric and topological notions of $\mathbb{C}^n$ and $\mathbb{R}^n \times \mathbb{R}^n$ coincide.

Definition 1.2. : Let $p \in \mathbb{N}$, for $z = (z_1, z_2, \dots, z_n) \in \mathbb{C}^n$, we have the following norms on $\mathbb{C}^n$:

$$\|z\|_\infty := max_{1\leq j\leq n}|z_j|$$

and

$$\|z\|_p := (\sum_{j=1}^{n} |z_j|^p)^{\frac{1}{p}}$$

here $\|.\|_\infty$ is called the **maximum norm**, $\|.\|_p$ is called the p-**norm**.

Definition 1.3. : Two norms $\|.\|_1, \|, \|_2$ on a vector space V are called **equivalent** if there are constants $c, c' \in \mathbb{R}_+$ such that

$$c\|x\|_1 \leq \|x\|_2 \leq c'\|x\|_1$$

for $\forall x \in V$.

Proposition 1.1. : *On a finite-dimensional vector space V over $\mathbb{R}$ or $\mathbb{C}$, all norms are equivalent.*

Proof: Since equivalence of norms is an equivalence relation, we just need to show that an arbitrary norm $\|.\|$ on V is equivalent to the Euclidian norm (2).

Let $\{b_1, b_2 \ldots, b_n\}$ be a basis of V and let $M := max\{\|b_1\|, \|b_2\|, \ldots, \|b_n\|\}$. For $\forall x \triangleq \sum_{j=1}^n \alpha_j b_j \in \mathbb{C}$ with $\alpha_j \in \mathbb{C}$. By triangle inequality and Holder Inequality, we get

$$\|x\| = \|\sum_{j=1}^n \alpha_j b_j\| \leq (\sum_{j=1}^n |\alpha_j|^2)^{\frac{1}{2}} (\sum_{j=1}^n \|b_j\|^2)^{\frac{1}{2}} \leq \|x\|_2 \sqrt{n} M$$

Since $|\|x\| - \|y\|| \leq \|x - y\|$, we know that norm is a continuous map, hence on the compact unit sphere

$$S := \{x \in V | \ \|x\|_2 = 1\}$$

$\|.\|$ attains a minimum $s \geq 0$ on S. Since $0 \notin S$, $\|x\| \neq 0$ on S, which means that $s>0$. For $\forall x \neq 0$, we have

$$\frac{x}{\|x\|_2} \in S$$

so we get

$$\|\frac{x}{\|x\|_2}\| \geq s>0$$

i.e., $\|x\| \geq s\|x\|_2$. Thus we have

$$s\|x\|_2 \leq \|x\| \leq \sqrt{n}M\|x\|_2$$

which shows that any norm $\|.\|$ is equivalent to the Euclidan norm on V. $\square$

Example 1.1. : On infinite-dimensional vector space, not all norms are equivalent. Consider the space $C^1[0,1]$, then we can define two norms on $C^1[0,1]$ as follows:

$$\|f\|_\infty := \sup_{x\in[0,1]}|f(x)|$$

and

$$\|f\|_{C^1} := \|f\|_\infty + \|f'\|_\infty$$

Now, take $f(x) = x^n$, then we have

$$\|f\|_\infty = \sup_{x\in[0,1]}|x^n| = 1$$

$$\|f\|_{C^1} = \sup_{x\in[0,1]}|x^n| + \sup_{x\in[0,1]}|nx^{n-1}| = 1 + n$$

Since $n \in \mathbb{N}$ is arbitraty, so there is no constant $c>0$ such that

$$\|f\|_{C^1} \leq c\|f\|_\infty$$

for all $f \in C^1[0,1]$.

Definition 1.4. : Let E be a real vector space. A subset $C \subseteq E$ is called **convex** if for $\forall x, y \in C$, we have $tx + (1-t)y \in C$ where $t \in [0,1]$.

Definition 1.5. : Let $M \subseteq E$ be an subset of E. The **convex hull of** M is the intersection of all convex subet of E containing M, which is denoted by $conv(M)$.

Example 1.2. : Let $r \in \mathbb{R}_+$ and $a \in \mathbb{C}^n$. The set

$$B_r^n(a) := \{z \in \mathbb{C}^n | \, \|z-a\| < r\}$$

is called the n-dimensional open ball centered at a and radius r with respect to the norm $\|.\|$. It is a convex set. Indeed, for $\forall z, w \in B_r^n(a)$ and $t \in [0,1]$, by the triangle inequality, we get

$$\|tz + (1-t)w - a\| \leq t\|z-a\| + (1-t)\|w-a\| < tr + (1-t)r = r$$

i.e., $tz + (1-t)w \in B_r^n(a)$. The closed ball $\overline{B_r^n(a)}$ is defined by

$$\overline{B_r^n(a)} := \{z \in \mathbb{C}^n | \, \|z-a\| \leq r\}.$$

Definition 1.6. : Let $r = (r_1, r_2, \ldots, r_n) \in \mathbb{R}_+^n$ and $a \in \mathbb{C}^n$.
The set

$$P_r^n(a) := \{z \in \mathbb{C}^n | \, |z_j - a_j| < r_j, \forall j = 1, 2, \ldots, n\}$$

is called the open **polycylinder** with center a and polyradius r.
The set

$$T_r^n(a) := \{z \in \mathbb{C}^n | \, |z_j - a_j| = r_j, \forall j = 1, 2, \ldots, n\}$$

is called the **polytorus** with center a and polyradius r. If $r_j = 1$ and $a = 0$, then it is called the **unit polytorus** and denoted by $\mathbb{T}^n$.

Remark 1.1. *: The polycylinder is the Cartesian product of n open discs in $\mathbb{C}$, thus it is also called* ***polydisc*** *and $P_r^n(a)$ is also convex.*

By topology, we have the following result

Proposition 1.2. *: Let C be a convex subset of $\mathbb{C}^n$. Then C is simply-connected.*

Recall the defintion of connnectedness and domains.

Definition 1.7. : Let X be a topological space.
(1) The space X is called **connected** if X cannot be represented as the disjoint union of two nonempty open subsets of X, i.e., if A, B are open subsets such that $A \cap B = \phi, A \neq \phi$ and $X = A \cup B$, then $B = \phi$ and $A = X$.
(2) An open and connected subset $D \subseteq X$ is called a **domain**.

We have equivalent characterizations of connectedness sets stated in the following lemma

Lemma 1.1. *: Let X be a topological space and $D \subseteq X$ an open subset in X. Then TFAE:*
(1) The set D is a domain;
(2) If $A \neq \phi$ is a subset of D which is both open and closed in D, then $A = D$;
(3) Every locally constant function $f : D \to \mathbb{C}$ is constant.

Proof: (1)$\Rightarrow$(2): Let $B := D \setminus A$, hence B is open in D and $A \cap B = \phi, D = A \cup B$. Since D is connected and $A \neq \phi$, thus $A = D$ and $B = \phi$;

(2)$\Rightarrow$(3): Let $c \in D$ and define $A := f^{-1}(\{f(c)\})$. Since $\mathbb{C}$ is a Hausdorff space and f is continuous (since f is locally constant), thus we know that $\{f(c)\}$ is closed in $\mathbb{C}$ and A is closed in D. $c \in A$, $A \neq \phi$. Let $p \in A$, then there exists an open neighborhood U of p such that $f(x) = f(p) = f(c)$ for $\forall x \in U$, i.e., $U \subseteq A$, which implies that A is open in D. Hence $A = D$ and $f : D \to \mathbb{C}$ is constant;

(3)$\Rightarrow$(1): If $D = A \cup B$, where A, B are open subsets of D and $A \cap B = \phi$, then

$$f : D \to \mathbb{C}$$

$$z \mapsto 1 \ (\forall z \in A)$$

$$z \mapsto 0 \ (\forall z \in B)$$

defines a locally constant function but not constant on D. $\square$

Remark 1.2. *: In the one-variable complex analysis, by Riemann Mapping Theorem, all simply-connected domains in $\mathbb{C}$ are biholomorphically equivalent to either $\mathbb{C}$ or the unit disc. But it is not true in the multivariable case. It will be shown that even the two natural generalizations of the unit disc, i.e., the unit ball and the unit polycylinder, are not biholomorphically equivalent.*

2 Holomorphic Functions in Several Complex Variables

Definition 2.1. : Let $U \subseteq \mathbb{C}^n$ be an open subset, $f : U \to \mathbb{C}^m$, $a \in U$ and $\|.\|$ is an arbitrary norm on $\mathbb{C}^n$.
(1) The function f is called **complex differentiable at** a if for $\forall\varepsilon>0$, $\exists\delta = \delta(\varepsilon, a)>0$ and a $\mathbb{C}$-linear map

$$Df(a) : \mathbb{C}^n \to \mathbb{C}^m$$

such that for all $z \in U$ with $\|z - a\|<\delta$, the inequality

$$\|f(z) - f(a) - Df(a)(z - a)\| \leq \varepsilon\|z - a\|$$

holds. If $Df(a)$ exists, it is called the **complex derivative** of f at a.
(2) f is called **holomorphic** on U if f is complex differentiable at all $a \in U$.
(3) Define the set

$$\mathcal{O}(U, \mathbb{C}^n) := \{f : U \to \mathbb{C}^m | \ f \text{ is holomorphic}\}$$

particularly, if $m = 1$, we denote

$$\mathcal{O}(U) := \mathcal{O}(U, \mathbb{C})$$

it is called the set of holomorphic functions on U.

Remark 2.1. *: Since all norms on $\mathbb{C}^n$ are equivalent, this definition is independent of the choice of the norm.*

Proposition 2.1. *:(1) If f is complex differentiable at $a \in U$, then f is continuous at a.*
(2) The derivative $Df(a)$ is unique if it exists.
(3) The set $\mathcal{O}(U, \mathbb{C}^m)$ is a $\mathbb{C}$-vector space and we have

$$D(\lambda f + \mu g)(a) = \lambda Df(a) + \mu Dg(a)$$

for $\lambda, \mu \in \mathbb{C}$ and $f, g \in \mathcal{O}(U, \mathbb{C}^m)$.
(4) (Chain Rule) Let $U \subseteq \mathbb{C}^n$ and $V \subseteq \mathbb{C}^m$ be open sets $a \in U$, if $f \in \mathcal{O}(U, V)$ and $g \in \mathcal{O}(V, \mathbb{C}^k)$, then $g \circ f \in \mathcal{O}(U, \mathbb{C}^k)$ and

$$D(g \circ f)(a) = D(g)(f(a)) \circ Df(a).$$

(5) Ket $U \subseteq \mathbb{C}^n$ be an open subset, the map

$$f = (f_1, f_2, \ldots, f_m) : U \to \mathbb{C}^m$$

is holomorphic $\Leftrightarrow$ all components f_i are holomorphic functions on U.
(6) $\mathcal{O}(U)$ is a $\mathbb{C}$-algebra. If $f, g \in \mathcal{O}(U)$ and $g(z) \neq 0$ for $\forall z \in U$, then $\frac{f}{g} \in \mathcal{O}(U)$.

Proof: Analogous to the real variable case. □

Example 2.1. : Let $U \subseteq \mathbb{C}^n$ be an open subset and $f : U \to \mathbb{C}$ is a locally constant function. Then

f is holomorphic and $Df(a) = 0$ for all $a \in U$.

Proof: For $\forall a \in U$ and $\forall \varepsilon > 0$. Sice f is locally constant, $\exists \delta > 0$ such that

$$\|f(z) - f(a)\| = 0 \leq \varepsilon \|z - a\|$$

for all $z \in U$ with $\|z - a\| \leq \delta$. i.e., f is holomorphic with $Df(a) = 0$ for all $a \in U$. $\square$

Example 2.2. : For every $k = 1, 2, \ldots, n$, the projection

$$pr_k : \mathbb{C}^n \to \mathbb{C}$$

$$(z_1, z_2, \ldots, z_n) \mapsto z_k$$

is holomorphic and $Dpr_k(a) = a_k$ for all $a = (a_1, a_2, \ldots, a_n) \in \mathbb{C}^n$.

Proof: Let $\varepsilon > 0$ and $a \in U$, then

$$|pr_k(z) - pr_k(a) - (z_k - a_k)| = 0 \leq \varepsilon \|z - a\|$$

for all $z \in \mathbb{C}^n$. $\square$

Example 2.3. : The complex subalgebra $\mathbb{C}[z_1, z_2, \ldots, z_n]$ of $\mathcal{O}(\mathbb{C}^n)$ generated by the constants and projections is called the **algebra of polynomials**. Its elements are sums of the form

$$\sum_{\alpha \in \mathbb{N}^n} c_\alpha z^\alpha$$

where $c_\alpha \neq 0$ only for finitely many $c_\alpha \in \mathbb{C}$ and for $z \in \mathbb{C}^n$ and $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in \mathbb{N}^n$ we use the notation

$$z^\alpha := z_1^{\alpha_1} z_2^{\alpha_2} \ldots z_n^{\alpha_n}.$$

The **degree** of the polynomial

$$p(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha z^\alpha \ (c_\alpha = 0 \text{ for almost all } \alpha)$$

is defined as

$$\deg p := max\{\alpha_1 + \alpha_2 + \cdots + \alpha_n |\ \alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n) \in \mathbb{N}^n, c_\alpha \neq 0\}.$$

3 Basic Properties of Holomorphic Functions

Lemma 3.1. *: Let $U \subseteq \mathbb{C}^n$ be open susbet, $a \in U$, $f \in \mathcal{O}(U)$ and $b \in \mathbb{C}^n$. $V := V_{a,b;U} := \{t \in \mathbb{C}|\ a + tb \in U\}$. Then V is open in $\mathbb{C}$, $0 \in V$ and the funciton*

$$g_{a,b} : V \to \mathbb{C}$$

$$t \mapsto f(a + tb)$$

is holomorphic.

Proof: It's clear that $0 \in V$. If $b = 0$, then $V = \mathbb{C}$. Now, suoopse $b \neq 0$, for $t_0 \in V$, then $z_0 := a + t_0 b \in U$. Since U is open, $\exists \varepsilon>0$ such that $B_\varepsilon(z_0) \subseteq U$. Let $z_t := a + tb$, then

$$\|z_0 - z_t\| \le \|b\| |t_0 - t|$$

for all t with $|t_0 - t| < \frac{\varepsilon}{\|b\|}$, which means that

$$B_{\frac{\varepsilon}{\|b\|}}(t_0) \subseteq V,$$

which implies that V is open. Since $g_{a,b}$ is the composition of the affine linear map $t \mapsto a + tb$ and the holomorphic function f, it's clear that $g_{a,b}$ is holomorphic. □

We have analogous of the following results from the one-variable dimensional theory.

Liouville's Theorem

Proposition 3.1. *: Every bounded holomorphic function*

$$f : \mathbb{C}^n \to \mathbb{C}$$

is constant.

Proof: Let $a, b \in \mathbb{C}^n$. The fuction $g_{a,b-a}$ from the lemma above is holomorphic on $\mathbb{C}$, which satisfies

$$g_{a.b-a}(0) = f(a), g_{a,b-a}(1) = f(b)$$

and

$$g_{a,b-a}(\mathbb{C}) \subseteq f(\mathbb{C}^n).$$

Since f is bounded, $g_{a,b-a}$ is bounded. By the one-dimensional version of Liouville's Theorem, $g_{a,b-a}$ is constant, which shows that $f(a) = f(b)$ and hence f is constant. □

Identity Theorem

Proposition 3.2. *: Let $D \subseteq \mathbb{C}^n$ be a domain, $a \in D, f \in \mathcal{O}(D)$ such that $f = 0$ near a.(i.e.,there is an open heighbourhood of a on which f is zero) Then f is zero function.*

Proof: Let

$$U := \{z \in D |\ f = 0 \text{ near } z\}$$

hence $a \in U$. U is closed in D, indeed, if f is the zero map then $U = D$, or otherwise, by the continuity of f, for every $z \in D \setminus U$, there exists a heighborhood W on which f does not vanish, i.e, $W \subseteq D \setminus U$.

For $\forall c \in U \cap D$, there is a polyradius $r \in \mathbb{R}^n_+$ such that the polycylinder $P_r(c) \subseteq D$ and $P_r(c) \cap U \neq \phi$. Take $z \in P_r(c)$ and take $w \in P_r(c) \cap U$, by the lemma above, we know that $V_{w,z-w;D}$ is open in $\mathbb{C}$ and because $P_r(c)$ is convex, we have $[0,1] \subseteq V_{w,z-w;D}$. Since f vanishes near w, there exists an open and connected neighborhood $W \subseteq \mathbb{C}$ of $[0,1]$ on which $g_{w,z-w}$ vanishes, i.e., $P_r(c) \subseteq U$ and hence U is open in D.

Since D is connected, thus the only nonempty open and closed subset of D is D itself, which means that $U = D$. Thus $f = 0$ on D. □

Open Mapping Theorem

Proposition 3.3. *: Let $D \subseteq \mathbb{C}^n$ be a domain, $U \subseteq D$ an open subset and $f \in \mathcal{O}(D)$ a non-constant function. Then $f(U)$ is open, i.e., every holpmorphic function is an open mapping. In particular, $f(D)$ is a domain in $\mathbb{C}$.*

Proof: By topology, since f is continuous and D is connected, we know that $f(D)$ is connected. We show that $f(U)$ is open:

Let $b \in f(U)$, hence $\exists a \in U$ such that $b = f(a)$. Since U is open, there exists a polycylinder $P_r(a) \subseteq U$. By the Identity Theorem, f is not constant on $P_r(c)$, since otherwise f would be constant on all of $P_r(a)$. This implies that there is some $w \in \mathbb{C}^n, w \neq 0$ such that $g_{a,w}$ from the lemma above is not constant on $V := V_{a,w;P_r(a)}$. By the one-dimensional case of open mapping theorem, we know that $g_{a,w}(V)$ is an open neighborhood of $b = f(a)$. Because

$$b \in g_{a,w}(V) \subseteq f(P_r(a)) \subseteq f(U),$$

$f(U)$ is a neighborhood of b. Since b is arbitrary, $f(U)$ is open in $\mathbb{C}$. □

Maximum Modulus Theorem

Proposition 3.4. *: If $D \subseteq \mathbb{C}^n$ is a domain, $a \in D$ and $f \in \mathcal{O}(U)$ such that $|f|$ has a local maximum at a, then f is constant.*

Proof: Since $|f|$ has a local maximun at $a \in U$, hence $\exists B_r(a) \subseteq D$ such that $|f(a)|$ is the maximum of $|f(z)|$ on $B_r(a)$. If f is non-constant, then by Open Mapping Theorem, we know that $f(B_r(a))$ is open in $\mathbb{C}$, particularly, $f(a)$ is an interior point of $f(B_r(a))$, which means that $|f(a)|$ is not the maximum. (contradiction!) Thus f must be constant. □

By Maximum Modulus Theorem, we have the following corollary

Maximum Modulus Principle for Bounded Domains

Corollary 3.1. : *Let $D \subseteq \mathbb{C}^n$ be a bounded domain and $f : \overline{D} \to \mathbb{C}$ is a continuous function, whose restriction to D is holomorphic. Then $|f|$ attains a maximum on the boundary ∂D.*

Proof: $\overline{D}$ is compact since D is bounded in $\mathbb{C}^n$. Thus the continuous real-valued function $|f|$ attains a maximum in a point $p \in \overline{D}$. If $p \in \partial D$, it's done. If $p \in D$, by Maximum Mudulus Theorem, we know that f is constant on $\overline{D}$ and thus $|f|$ also attains a maximum on ∂D. $\square$

In the one-dimensional case of Identity Theorem, it is sufficient to know the values of a holomophic function on a subset of a domain, which has an accumualtion point. But it is not true for muti-variable case.

Example 3.1. : The holomorphic function

$$f : \mathbb{C}^2 \to \mathbb{C}$$

$$(z, w) \mapsto zw$$

is not identitcally zero. But f vanishes on the subsets $\mathbb{C} \times \{0\}$ and $\{0\} \times \mathbb{C}$ of $\mathbb{C}^2$, which clearly have accumulation points in $\mathbb{C}^2$.

4 Partially Holomorphic Functions and Cauchy-Riemann Equations

Definition 4.1. : Let $U \subseteq \mathbb{C}^n$ be an open subset, $a = (a_1, \dots, a_n) \in U$ and $f : U \to \mathbb{C}$. For $j = 1, 2, \dots, n$, define

$$U_j := \{z \in \mathbb{C} |\ (a_1, \dots, a_{j-1}, z, a_{j+1}, \dots, a_n) \in U\}$$

and

$$\hat{f}_j : U_j \to \mathbb{C}$$

$$z \mapsto f(a_1, \dots, a_{j-1}, z, a_{j+1}, \dots, a_n).$$

f is called **partially holomorphic** on U if all $\hat{f}_j$ are holomorphic.

A function f holomorphic on an open subset $U \subseteq \mathbb{C}^n$ can be considered as a totally differentiable function of $2n$ real variables. We define

$$\mathcal{C}^k(U) := \{f : U \to \mathbb{C} |\ f \text{ is } k \text{ times } \mathbb{R} - \text{differentiable}\}$$

where $k = 0$ denotes the continuous functions and $\mathcal{C}^0(U) \triangleq \mathcal{C}(U)$.

Let $a \in U$ and $f \in \mathcal{C}^1(U)$, then there is an $\mathbb{R}$-linear function

$$d_a f : \mathbb{R}^{2n} \to \mathbb{C}$$

called the **real differential** of f at a such that

$$f(z) = f(a) + d_a f(z - a) + O(\|z - a\|^2).$$

Comparing this to Definition2.1 we can say that f is $\mathbb{C}$-linear at a iff $d_a f$ is $\mathbb{C}$-linear. So we look closer at the relationship between $\mathbb{R}$-linear and $\mathbb{C}$-linear functions of complex vector spaces.

Lemma 4.1. : *Let V be a vector space over $\mathbb{C}$ and $V^\#$ be its algebraic dual, i.e.,*

$$V^\# := \{f : V \to \mathbb{C} |\ f \textit{ is } \mathbb{C} - \textit{linear}\}.$$

Furthermore, we define

$$\overline{V}^\# := \{f : V \to \mathbb{C} |\ f \textit{ is } \mathbb{C} - \textit{antilinear}\}$$

and

$$V_\mathbb{R}^\# := \{f : V \to \mathbb{C} |\ f \textit{ is } \mathbb{R} - \textit{linear}\}.$$

Then $V_\mathbb{R}^\#$ is a complex vector space, $V^\#$ and $\overline{V}^\#$ are subspaces of $V_\mathbb{R}^\#$ and we have the direct decomoosition

$$V_\mathbb{R}^\# = V^\# \oplus \overline{V}^\#.$$

Proof: It's clear that $V_{\mathbb{R}}^{\#}$ is a complex vector space.

Let $\mu \in V^{\#} \cap \overline{V}^{\#}$, since μ is both $\mathbb{C}$-linear and $\mathbb{C}$-antilinear, so for $\forall z \in V$, we have

$$\mu(iz) = i\mu(z) = -i\mu(z)$$

which implies that $\mu = 0$ and hence $V^{\#} \cap \overline{V}^{\#} = \{0\}$;

For $\forall \mu \in V_{\mathbb{R}}^{\#}$, define

$$\mu_1(z) := \frac{1}{2}(\mu(z) - i\mu(iz))$$

$$\mu_2(z) := \frac{1}{2}(\mu(z) + i\mu(iz))$$

it's easy to show that μ_1 and μ_2 are both $\mathbb{R}$-linear and $\mu = \mu_1 + \mu_2$. Since

$$\mu_1(iz) = \frac{1}{2}(\mu(iz) - i\mu(-z)) = i\frac{1}{2}(\mu(z) - i\mu(iz)) = i\mu_1(z)$$

$$\mu_2(iz) = \frac{1}{2}(\mu(iz) + i\mu(-z)) = -i\frac{1}{2}(\mu(z) + i\mu(iz)) = -i\mu_2(z)$$

so we get $\mu_1 \in V^{\#}$a and $\mu_2 \in \overline{V}^{\#}$. Thus $V_{\mathbb{R}}^{\#} = V^{\#} \oplus \overline{V}^{\#}$. $\square$

Apply this lemma to the case that $V := \mathbb{C}^n = \mathbb{R}^n + i\mathbb{R}^n$. Let $w = u + iv \in V$. For $j = 1, \ldots, n$, consider the linear functionals

$$dx_j, dy_j : \mathbb{C}^n \to \mathbb{C},\ dx_j(w) := u_j, dy_j(w) := v_j.$$

It's clear that $dx_j, dy_j \in V_{\mathbb{R}}^{\#}$ and $dx_j(iw) = -v_j, dy_j(iw) = u_j$.

Now, define

$$dz_j := dx_j + idy_j$$

$$d\overline{z_j} := dx_j - idy_j$$

then we get

$$dz_j(w) = u_j + iv_j = w_j = dx_j(w) - idx_j(iw)$$

$$d\overline{z_j}(w) = u_j - iv_j = \overline{w_j} = dx_j(w) + idx_j(iw)$$

It's easy to check that

$$dz_j \in V^{\#}, d\overline{z_j} \in \overline{V}^{\#};$$

Given the linear equation

$$\sum_{j=1}^{n} \alpha_j dz_j = 0$$

where $\alpha_j \in \mathbb{C}$, apply this to the canonical basis $\{e_1, e_2, \ldots, e_n\}$ of $\mathbb{C}^n$, we can get $\alpha_j = 0$, (Note that $\dim_{\mathbb{C}}\mathbb{C}^n = n$ and we have the canonical $\mathbb{C}$-basis $\{e_1 = (1, 0, \ldots, 0), e_2 = (0, 1, \ldots, 0), \ldots, e_n = (0, 0, \ldots, 1)\}$) which means that $\{dz_1, dz_2, \ldots, dz_n\}$ is $\mathbb{C}$-linearly independent. By linear algebra, we know that $\dim V^{\#} = \dim V = \dim_{\mathbb{C}}\mathbb{C}^n = n$, thus $\{dz_1, dz_2, \ldots, dz_n\}$ forms a basis of $V^{\#}$.

By the same discussion, we also know that $\{d\overline{z_1}, d\overline{z_2}, \ldots, d\overline{z_n}\}$ forms a basis of $\overline{V}^{\#}$.

So by linear algebra, we know that $\{dz_1, dz_2, \ldots, dz_n, d\overline{z_1}, d\overline{z_2}, \ldots, d\overline{z_n}\}$ forms a $\mathbb{C}$-basis of $V_{\mathbb{R}}^{\#}$. This leads to the following representation of the real differential $d_a f$:

$$d_a f \triangleq \sum_{j=1}^{n} (\alpha_j(f,a)dz_j + \beta_j(f,a)d\overline{z_j})$$

where the cofficients $\alpha_j(f,a), \beta_j(f,a) \in \mathbb{C}$.

Notation: Denote

$$\partial_j f(a) := \frac{\partial f}{\partial z_j}(a) := \alpha_j(f,a);\ \overline{\partial_j} f(a) := \frac{\partial f}{\partial \overline{z_j}}(a) := \beta_j(f,a)$$

Hence we have the following definition

Definition 4.2. : The linear functional

$$\partial_a f := \sum_{j=1}^{n} \frac{\partial f}{\partial z_j}(a)dz_j : \mathbb{C}^n \to \mathbb{C}$$

$$w \mapsto \sum_{j=1}^{n} \frac{\partial f}{\partial z_j}(a)w_j$$

is called the **complex differential** of f at a. The antilinear functional

$$\overline{\partial_a} f := \sum_{j=1}^{n} \frac{\partial f}{\partial \overline{z_j}}(a)d\overline{z_j} : \mathbb{C}^n \to \mathbb{C}$$

$$w \mapsto \sum_{j=1}^{n} \frac{\partial f}{\partial \overline{z_j}}(a)\overline{w_j}$$

is called the **comlex-conjugate differential** of f at a.

So by these definitions, we get the decomposition of the real differential

$$d_a f = \partial_a f + \overline{\partial_a} f \in (\mathbb{C}^n)^{\#} \oplus \overline{\mathbb{C}^n}^{\#} \ \ (i.e., d_a = \partial_a + \overline{\partial_a})$$

Hence these results can be summarized in the following theorem

Cauchy-Riemann

Theorem 4.1. *: Let $U \subseteq \mathbb{C}^n$ be an open subset and $f \in \mathcal{C}^1(U)$. Then TFAE:*
(1) The funtion f is holomorphic on U;
(2) For $\forall a \in U$, we have $d_a f \in (\mathbb{C}^n)^{\#}$, i.e., $d_a f$ is $\mathbb{C}$-linear;
(3) For $\forall a \in U$, the equation $\overline{\partial_a} f = 0$ holds;
(4) For $\forall a \in U$, the function f satisfies the Cauchy-Riemann differential equations

$$\frac{\partial f}{\partial \overline{z_j}}(a) = 0$$

for all $j = 1, 2, \ldots, n$ on U.